

Multiplications and Division of Fractions 3

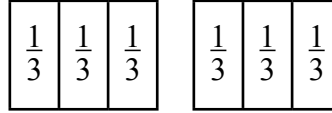


2 means 2 whole ones and can be written as $\frac{2}{1}$

$\frac{1}{2}$ and $\frac{2}{1}$ are called the **inverse** of each other.

A). E.g.

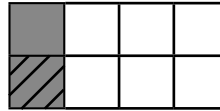
$2 \div \frac{1}{3}$ can be shown as



So $2 \div \frac{1}{3} = \underline{6}$

E.g.

$\frac{1}{4} \div 2$ means $\frac{1}{4}$ cut in two.



So $\frac{1}{4} \div 2 = \underline{\frac{1}{8}}$

By drawing out appropriate diagrams, or otherwise, find

- 1). $3 \div \frac{1}{4}$
- 2). $4 \div \frac{1}{2}$
- 3). $4 \div \frac{1}{3}$
- 4). $2 \div \frac{1}{5}$
- 5). $3 \div \frac{1}{5}$
- 6). $\frac{1}{3} \div 2$
- 7). $\frac{1}{4} \div 3$
- 8). $\frac{1}{6} \div 2$
- 9). $\frac{1}{5} \div 4$
- 10). $\frac{1}{3} \div 5$
- 11). $6 \div \frac{1}{5}$
- 12). $7 \div \frac{1}{6}$
- 13). $10 \div \frac{1}{4}$
- 14). $8 \div \frac{1}{9}$
- 15). $9 \div \frac{1}{10}$
- 16). $\frac{1}{7} \div 5$
- 17). $\frac{1}{6} \div 7$
- 18). $\frac{1}{8} \div 8$
- 19). $\frac{1}{11} \div 9$
- 20). $\frac{1}{9} \div 12$
- 21). Write a rule how to find the answer **without** drawing a diagram.

B). To divide a number by a fraction, **multiply** the number by the **inverse** of the fraction.

E.g. $5 \div \frac{2}{3} = 5 \times \frac{3}{2} = \frac{15}{2} = 7\frac{1}{2}$

E.g. $8 \div \frac{3}{5} = 8 \times \frac{5}{3} = \frac{40}{3} = 13\frac{1}{3}$

Find

- 1). $7 \div \frac{2}{3}$
- 2). $3 \div \frac{2}{5}$
- 3). $7 \div \frac{5}{6}$
- 4). $4 \div \frac{3}{8}$
- 5). $5 \div \frac{2}{9}$
- 6). $8 \div \frac{3}{4}$
- 7). $6 \div \frac{5}{8}$
- 8). $7 \div \frac{2}{3}$
- 9). $5 \div \frac{3}{10}$
- 10). $4 \div \frac{7}{10}$
- 11). $9 \div \frac{2}{3}$
- 12). $5 \div \frac{3}{4}$
- 13). $8 \div \frac{5}{6}$
- 14). $9 \div \frac{7}{8}$
- 15). $10 \div \frac{3}{4}$
- 16). $13 \div \frac{2}{5}$
- 17). $16 \div \frac{3}{8}$
- 18). $14 \div \frac{5}{6}$
- 19). $21 \div \frac{2}{3}$
- 20). $23 \div \frac{4}{7}$

